

codingmafias

Port Flow

Problem

90% of all global trade is transported by sea.

During peak congestion (like the 2021-2022 crisis), ships waited up to **41 days** at anchorages outside ports like Los Angeles and Long Beach.

The daily operating cost of a large container ship sitting idle is roughly **\$50,000 to \$150,000 per day**.



Why Existing Approaches Fall Short

Manual & Reactive: 80% of port scheduling globally is still done manually via spreadsheets and whiteboards.

"First Come, First Served"

(FCFS): Traditional queuing ignores vessel capacity and unloading time, leading to cascading delays.

Siloed Visibility: Ships spend 15+ days crossing the ocean completely blind to the real-time congestion forming at their destination port.



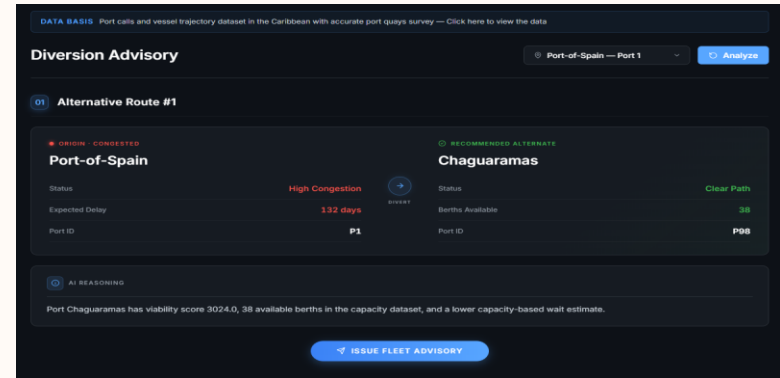
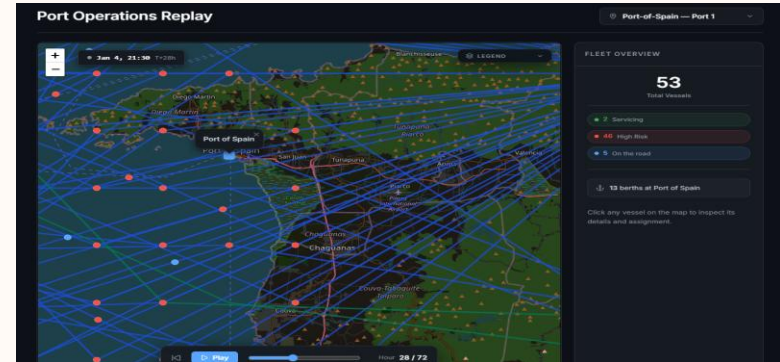
Solution

Proactive, not Reactive: We predict gridlock days before ships arrive.

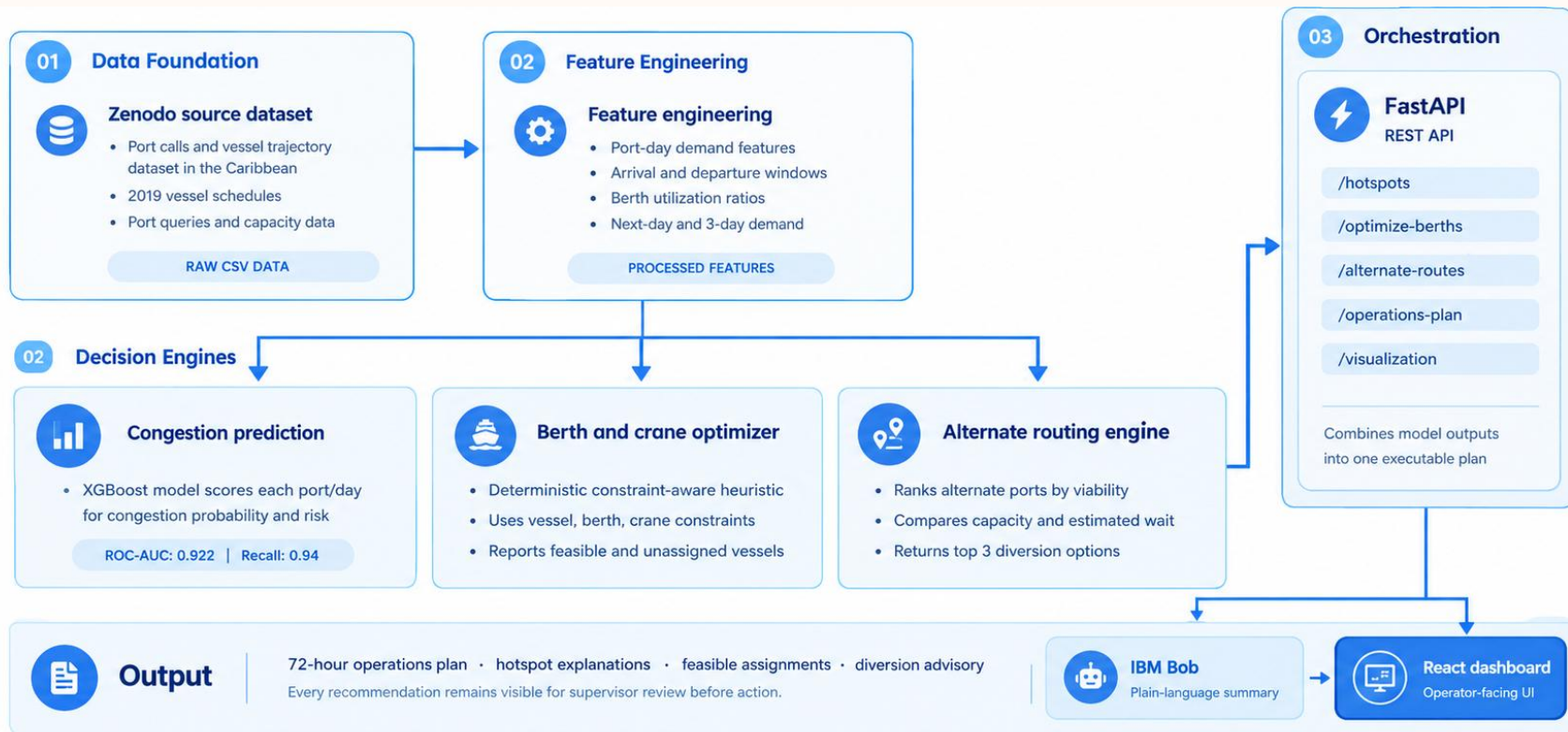
Heuristic Scheduling: Mathematically optimizing physical constraints, replacing the flawed "First Come, First Served" model.

Automated Detours: Re-routing ships to alternative ports dynamically.

Human-in-the-Loop: Translating complex AI data into plain-English instructions.



Architecture



Demo

DATA BASIS Port calls and vessel trajectory dataset in the Caribbean with accurate port quays survey — [Click here to view the data](#)

72-Hour Operations Plan Refresh

EXECUTIVE SUMMARY

HIGH congestion risk at Port 1. 88 vessels are in the horizon; 88 are assigned and 0 remain unassigned (100% assignment coverage), with 1970.9 hours average wait. Prioritize high-risk vessels and review the next feasible berth plan. Keep Chaguaramas as an alternative.

NETWORK RISK

HIGH

CRITICAL SUPERVISOR ACTIONS (24 PENDING)

PRIORITY	TASK DESCRIPTION	DEADLINE	STATUS
P0	Review Vessel 89879 at Berth 9; estimated waiting time is 160004 minutes.	4 Jan, 02:28 am	Review
P0	Review Vessel 95365 at Berth 9; estimated waiting time is 159325 minutes.	4 Jan, 01:41 am	Review
P0	Review Vessel 167067 at Berth 9; estimated waiting time is 159313 minutes.	4 Jan, 02:19 am	Review
P0	Review Vessel 91732 at Berth 9; estimated waiting time is 158434 minutes.	4 Jan, 12:58 am	Review
P0	Review Vessel 131588 at Berth 9; estimated waiting time is 158419 minutes.	4 Jan, 01:26 am	Review
P0	Review Vessel 88521 at Berth 9; estimated waiting time is 157058 minutes.	3 Jan, 09:29 pm	Review
P0	Review Vessel 16134 at Berth 9; estimated waiting time is 155785 minutes.	3 Jan, 09:27 pm	Review
P0	Review Vessel 80898 at Berth 9; estimated waiting time is 152903 minutes.	3 Jan, 09:26 pm	Review

Generative AI Shift

Reports: IBM Watsonx translates raw predictive math into plain-English summaries instantly.

Automated Action

Triage: Critical bottlenecks (P0 tasks) are automatically flagged and prioritized for supervisors.

Predictive Risk Alerts: Instantly surfaces high network congestion risks before they paralyze the port.

IBM technology integration

Subscription

Details about your subscription, including plan type, seats, Bobcoins, and add ons.

Account	bhavikapatel4298@gmail.com
Plan	Trial
Region	United States (East)

Monthly Bobcoin usage

Your budget resets every month and renews on October 14, 2026

Total spent ⓘ

12

Bobcoins

Budget ⓘ

50

Bobcoins

"We integrated the IBM Bob LLM into our Python backend as an automated logistics supervisor that analyzes real-time network congestion and algorithmic data. It translates complex port traffic scenarios into proactive, human-readable fleet diversion strategies and executive summaries for our operators."

```
○ (.venv) jinayshah@Jinays-MacBook-Air bob-ai-hackathon-codingmafias % uvicorn src.api.main:app --reload
INFO: Will watch for changes in these directories: ['/Users/jinayshah/Desktop/bob-ai-hackathon-codingmafias']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [49451] using StatReloader
INFO: Started server process [49453]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:57959 - "GET /api/v1/alternate-routes?vessel_id=demo-vessel&port_id=1 HTTP/1.1" 200 OK

[BOB-AI] Calling IBM Bob inference...
[BOB-AI] Prompt Length: 2450 characters
[BOB-AI] SUCCESS - Received response (430 characters)
[BOB-AI] Output Preview: Based on the current network congestion at Port 1...

INFO: 127.0.0.1:57974 - "GET /api/v1/operations-plan HTTP/1.1" 200 OK
```

```
# 3. Call IBM Bob inference
formatted_prompt = PROMPT.format(
    plan=json.dumps(plan, separators=(",", ":"))
)

try:
    result = subprocess.run(
        ["bob", "run", formatted_prompt],
        capture_output=True,
        text=True,
        timeout=15,
        env=os.environ.copy()
    )
    if result.returncode == 0 and result.stdout.strip():
        summary = result.stdout.strip()
        # Cache the generated summary
        SUMMARY_CACHE_FILE.write_text(summary, encoding="utf-8")
        return summary
except Exception:
    pass
```

Impact

- **Financial:** \$50k–\$150k daily idle cost per ship. Saving 2 days for 100 ships = **\$10,000,000+ saved.**
- **Environmental:** Zero-idle routing cuts thousands of tons of heavy bunker fuel emissions (CO₂, SO_x, NO_x) normally burned while anchored.
- **Operational:** 858,000 ships mathematically proven unassignable without proactive routing. AI prevents the supply chain "Bullwhip Effect".



Team



Team Lead – Bhumi Shah
Lead Architect & Optimization
Engineer
Heuristic Berth & Crane
Optimization



Bhavika Patel
Integrated IBM Watsonx
(Bob) to build the
Generative AI Explainability
Layer, UI/UX



Jinay Shah
Data Engineer & UI/UX
Designed the interactive React
Dashboard and engineered the
core data ingestion pipelines.



Yug Yadav
Machine Learning &
Backend Engineer
Developed the Predictive
Engine and architected the
Python backend.